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United States Patent	12388135
Kind Code	B2
Date of Patent	August 12, 2025
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Hybrid hydrogen power module

Abstract

An apparatus for powering trucks including a power module skid and supporting structure for fitting on a truck. The skid housing hydrogen fuel cell modules, battery sub packs, cooling means and cooling management, and integrated power electronics, to provide an electrical drive train of the truck with a constant high voltage DC power supply. An integrated system using renewable energy to reduce greenhouse gases using one or more trucks, in which an integrated system includes: means for providing renewable energy; means for using the renewable energy to synthesise hydrogen; means for storing the synthesised hydrogen. The integrated system includes hybrid hydrogen power modules fitted to each truck including hydrogen fuel cell modules and battery sub packs so that the battery sub packs and the battery sub packs are recharged by the hydrogen fuel cells.

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Appl. No.: 18/625294

Filed: April 03, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240266641 A1	Aug. 08, 2024

Foreign Application Priority Data

GB	1914406	Oct. 04, 2019
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Related U.S. Application Data

continuation parent-doc US 18545766 20231219 US 11955620 child-doc US 18625294
continuation parent-doc US 17425694 US 11894539 20240206 WO PCT/EP2020/077348
20200930 child-doc US 18545766

Publication Classification

Int. Cl.: **H01M16/00** (20060101); **B60L50/75** (20190101); **B60L53/22** (20190101); **B60L58/26** (20190101); **B60L58/33** (20190101); **B60L58/40** (20190101); **H01M8/04029** (20160101); **H01M8/0656** (20160101); **H01M10/613** (20140101); **H01M10/625** (20140101); **H01M10/663** (20140101)

U.S. Cl.:

CPC **H01M16/006** (20130101); **B60L50/75** (20190201); **B60L53/22** (20190201); **B60L58/26** (20190201); **B60L58/33** (20190201); **B60L58/40** (20190201); **H01M8/04029** (20130101); **H01M8/0656** (20130101); **H01M10/613** (20150401); **H01M10/625** (20150401); **H01M10/663** (20150401); B60L2200/36 (20130101); B60L2200/44 (20130101); B60L2210/10 (20130101); B60L2210/14 (20130101); H01M2220/20 (20130101); H01M2250/20 (20130101)

Field of Classification Search

CPC: B60L (50/75); B60L (53/22); B60L (58/26); B60L (58/33); B60L (58/40); B60L (2200/36); B60L (2200/44); B60L (2210/10); C25B (1/04); Y02E (60/36); Y02P (20/133); Y02T (90/14); B60Y (2200/142)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/545,766, filed Dec. 19, 2023, entitled “Hybrid Hydrogen Power Module,” now U.S. Pat. No. 11,955,620, which is a continuation of U.S. patent application Ser. No. 17/425,694, filed Jul. 23, 2021, entitled “Hybrid Hydrogen Power Module,” now U.S. Pat. No. 11,894,539, which is a National Phase Application under 35 U.S.C. § 371 of International Patent Application No. PCT/EP2020/077348, filed Sep. 30, 2020, entitled “Hybrid Hydrogen Power Module,” which claims priority to and the benefit of U.K. Patent Application No. GB 1914406.2, filed Oct. 4, 2019, entitled “Hybrid Hydrogen Power Module,” the entire disclosure of each of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention is directed towards the utilisation of energy to help power a truck, in particular for use on a truck in a mining environment. The invention is particularly directed towards the use of fuel cells on trucks to help eliminate greenhouse gas emissions on a site.

BACKGROUND

(2) Existing technology is entirely dependent on traditional fossil fuel sources. Trucks are powered

by diesel generators and the diesel is itself generated from the extraction and treatment of traditional fossil fuels. Mining plants are typically powered off the local grid which in most instances is largely or completely resourced by the burning of fossil fuel. There is an industry desire to reduce the dependence on fossil fuels and to provide a cost-effective alternative to diesel which alternative results in no or substantially no reduction in performance of the truck. An object of the present invention is therefore to provide apparatus to power trucks which does not rely on fossil fuels and which can both be retrofitted on to existing trucks in place of existing diesel generators, and can also be used in new purpose-built trucks. It is a further object of the invention to significantly reduce the carbon footprint of a mine and plant and the emission of greenhouse gases from a mine and associated trucks.

SUMMARY

(3) According to the present invention there is provided apparatus for powering trucks, said apparatus including a power module skid and supporting structure for fitting on a truck, said skid housing one or more hydrogen fuel cell modules, one or more battery sub packs, cooling means and cooling management, and integrated power electronics, to provide an electrical drive train of the truck with a constant high voltage DC power supply. By using the hybrid hydrogen power module including the hydrogen fuel cells and the battery sub packs, the trucks have a secure supply of energy throughout the day and night. The battery sub packs preferably recharge during regenerative braking of the trucks and also by drawing energy from the hydrogen fuel cells. Accordingly, there is no need to stop the trucks to recharge the batteries during the working day, or to remove the batteries for charging and replacing them with charged batteries. Both of these options are time consuming and result in the truck being out of use for significant proportions of time.

(4) The apparatus of the present invention is particularly effective when used on ultra heavy-duty trucks which may be used in a mine environment. These may be trucks which are over 100 tonnes, for example, or over 200 or 300 tonnes. Such ultra heavy-duty trucks require a considerable power input and are traditionally powered by heavy duty diesel generators. If they were to try to operate using batteries alone, they would require several batteries and such a set up would result in a reduction in the performance of the truck when compared to traditional diesel trucks, due in part to the weight of the batteries. The use of a hybrid system allows a significant reduction in the number of battery sub packs required and also provides a boost in power such that the performance can match or even exceed that of a traditional diesel driven truck, and at a similar weight.

(5) The skid may house any number of hydrogen fuel cell modules and any number of battery sub packs. This may vary depending on the environment in which the truck is being used and taking into account variables such as the nature of the terrain, the changes in elevation that the truck would have to travel, the overall distances travelled on a round trip, the ratio between travel uphill, downhill and on the flat, the nature and weight of the truck load, whether load is carried uphill, downhill or both and the climate (including, for example, the ambient temperature, humidity, wind speed, pressure). The flexibility of the apparatus of the present invention is important, so that optimal operation of the trucks can be achieved. In some embodiments, different skids having different combinations of hydrogen fuel cells and battery sub packs will be available and can be changed on trucks to take into account changes in operating conditions over time. In other embodiments, the configuration of a skid may be changed over time to increase or reduce the number of hydrogen fuel cells and/or battery sub packs, again to adapt to changes in the operating conditions.

(6) In an embodiment of the invention, there are equal numbers of hydrogen fuel cells and battery sub packs so that there is a 1:1 relationship between them. In other embodiments there are more hydrogen fuel cells than battery sub packs, for example in the range 10:1 to 2:1. In other alternative embodiments, there are fewer hydrogen fuel cells than battery sub packs, for example in the range 1:2 to 1:10.

(7) In an embodiment the cooling is provided, at least in part, by means of a multi-zone radiator

with integrated fans and coolant fluid loops. The cooling management ensures that the components are all efficiently and effectively cooled to maintain their temperatures within the operating range and such that the coolant is used as efficiently as it can be in the system.

(8) The apparatus of the present invention may produce a power output in excess of 0.5 MW, for example greater than 1 MW such as in the range 1 to 3 MW or 1.25 to 2.75 MW. The energy storage capacity of the apparatus may be in excess of 1 MWh, for example in the range 1 MWh to 5 MWh.

(9) The present invention also provides a method for powering trucks in a mine environment, in which a truck is provided with one or more hydrogen fuel cell modules and one or more battery sub packs and in which hydrogen is loaded on to the trucks and is used to generate power in the hydrogen fuel cells, said power being used both to run the truck operations and also to recharge the battery sub packs.

(10) The number of fuel cell modules and the number of battery sub-packs may be varied to control the overall battery energy storage capacity and therefore the power output as described above. The fuel cell modules may be arranged in parallel and be used simultaneously to provide power to a single DC link voltage which provides the power to the drive train of the truck.

(11) Additionally, there may be a DC/DC converter between each of the hydrogen fuel cell modules to boost the module's output voltage and to compensate for phases and output differences between the modules, to thereby enable a shared output to be regulated to provide a steady and consistent high voltage output. This configuration may provide substantial flexibility in the system and allow the power output to be controlled such that it is a steady and consistent high voltage output that may be regulated around voltage, current and/or power depending on the requirements of the truck and the environment in which it is being used.

(12) In an embodiment of the method, the battery sub packs recharge during operation by taking energy both from regenerative braking of the truck and from the hydrogen fuel cell.

(13) The present invention also extends to an integrated system for using renewable energy in a mine environment to reduce the emission of greenhouse gases, in which environment, mined material is transported around the mine using one or more trucks, in which integrated system there are: means for providing renewable energy; means for using the renewable energy to synthesise hydrogen; means for storing the synthesised hydrogen; wherein, the integrated system further includes hybrid hydrogen power modules fitted to each truck, said power modules including hydrogen fuel cell modules and battery sub packs, whereby the stored hydrogen is loaded on to the truck to feed the hydrogen fuel cell modules so that the truck is entirely powered by the combination of the hydrogen fuel cells and the battery sub packs and the battery sub packs are recharged by the hydrogen fuel cells during use such that they do not need to be removed or remotely charged.

(14) Such a hybrid energy system also provides a steady and consistent power supply for use in the trucks throughout the day and night. Use of renewable energy gives the user an independence from and resilience to fluctuations in energy prices which are dominated by the price of fossil fuels. The system also offers significant reductions in greenhouse gas emissions which helps to provide an environmental benefit to local communities (such as noise reduction and air quality improvements). It can also contribute towards a significant reduction in the carbon footprint of the mine operator.

(15) The source of renewable energy may be 100% renewable energy from a local grid system in locations where this is available. Alternatively, it may be local capture of renewable energy such as solar, wind, hydroelectric, geothermal or nuclear using any appropriate apparatus. This could, for example, be the use of photovoltaic cells to capture solar energy. In another embodiment, the invention could be put into effect using energy from a local grid system which is not obtained entirely from renewable sources but is, for example, predominantly from renewable sources such as at least 60%, or 70% or 80% or 90%.

(16) The mine environment may also include a plant where the mined material is processed, and

the capture of renewable energy is in excess of the demands of the plant and the excess energy is used to generate hydrogen which is then used in the powering of the trucks. The trucks may be fitted to include the following components: hydrogen storage tanks, battery packs, hydrogen fuel cell converters, fuel cell boosters, cooling apparatus, and control systems.

(17) The apparatus is optionally arranged to fit on a skid which can be retrofitted to existing trucks in place of the existing diesel power units. This allows the technology to be easily fitted to existing trucks by removing the skid with the diesel generator on it and replacing it with the new hybrid skid.

(18) The hydrogen may be synthesised by means of any suitable green hydrogen or carbon free process such as by the electrolysis of water.

(19) The invention also extends to a method for reducing the carbon footprint of a mining environment, in which the mining environment comprises a plant and a mine in which material is transported between the mine and the plant by means of one or more trucks, which method comprises: providing locally generated renewable energy; using the renewable energy to synthesise hydrogen; storing the hydrogen; operating the trucks by means of a hybrid hydrogen power module which includes hydrogen fuel cell modules and battery sub packs; in which the stored hydrogen is loaded on to the trucks and is used to generate power in the hydrogen fuel cells, said power being used to run the truck operations and also to recharge the battery sub packs.

(20) The hydrogen is preferably produced by utilising locally produced renewable energy. This can either be green energy taken from an existing grid, or could alternatively be through the capture of renewable energy from a source of renewable energy such as solar, wind, hydroelectric, geothermal or nuclear. This renewable energy can be used to generate hydrogen by splitting water using the process of electrolysis, as is known in the art. While it is recognised that this method of production may not be the most efficient way to generate hydrogen, if it is done using renewable energy then the effect on the overall carbon footprint of the operation is zero. The hydrogen produced this way would have a zero-carbon footprint as would the trucks then using this hydrogen (instead of diesel) to power the truck.

(21) Optional and preferred features from one aspect of the invention may equally apply to other aspects. Further benefits and advantages of the present invention will become apparent from the following description of embodiments exemplifying the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of a configuration of a power module skid according to an embodiment of the present invention;

(2) FIG. 2 is an electrical schematic of the power module of an embodiment of the present invention;

(3) FIG. 3 is schematic of the present invention being used on a truck in a mine environment; and

(4) FIG. 4 shows a graphical representation of the shift in energy for one cycle of the operation of the truck shown schematically in FIG. 3.

DETAILED DESCRIPTION

(5) Referring to FIG. 1, there is shown the integrated configuration of components on a skid **151** according to an embodiment of a hybrid hydrogen power module **150** according to the present invention. This skid module **151** can be readily fitted in to an existing truck in place of the diesel generator currently used. It could also be used on a new truck, constructed specifically to utilize the hybrid system. The module **150** consists of six major subsystem elements. Firstly, there is a power module skid **151** and supporting structure for carrying the components and enabling them to be configured for operation once the skid **151** has been fitted to the truck. Each skid **151** in this

embodiment includes eight hydrogen fuel cell modules **155** located in the middle of the skid **151**. These are fed by hydrogen from hydrogen storage tanks (not shown) which may be located at suitable locations on the truck. The position of these may vary depending on the equipment attached to the truck and the environment and terrain in which the truck is operating. Each fuel cell **155** will be fed from one or more hydrogen storage tank, preferably by several tanks so that operation of the fuel cell **155** is not interrupted as a storage tank empties.

(6) The skid **151** also includes a fuel cell coolant management system **156**. This may be arranged to be adjacent to the fuel cells **155**, for example towards the front of the skid **151**. By locating the coolant management system **156** close to the fuel cell modules **155**, there is minimum additional tubing and wiring and the cells **155** can be maintained in their optimum operating window. The use of a fluid loop necessitates intimate contact between the heat source and the fluid line and therefore the fluid lines must be integrated into the system design to provide effective heat dissipation and management.

(7) In the embodiment shown, the skid **151** holds eight battery sub-packs **154** such that there is one battery sub-pack **154** for each hydrogen fuel cell module **155**. The fuel cells **155** operate in parallel and can therefore all be used to charge the batteries **154**. The battery subpacks **154** all work together and can be considered to serve as one large effective battery pack. The batteries **154** are the heaviest components on the skid **151** and therefore, for best balance, they are located at the bottom of the skid **151** and some may be positioned under the fuel cell modules **155**.

(8) The components on the skid **151** are controlled by integrated power electronics which in the present example are located in a cabinet **153** towards the rear of the skid **151**. These are out of the way of the fuel cells **155** and can be positioned in any suitable location.

(9) At the front of the skid **151** is a power module radiator **152**. This integrated multi-zone thermal radiator **152** more efficiently manages the varying thermal needs of the power module components by providing a high local and well-connected thermal sink. This increases the efficiency of the cooling of the components on the skid **151**.

(10) The skid **151** may additionally include one or more fuel cell boost convertors (not shown) between the fuel cell modules **155** and the battery sub packs **154**. If present, there is preferably one fuel cell boost convertor for each fuel cell **155**. Such components help the power module **150** have a high level of flexibility of components, as illustrated in FIG. 2.

(11) As shown in FIG. 2, the hybrid power module **150** of the present invention has a significant flexibility of its configuration to reflect the overall “balance of plant”; that is, the architecture of components on the skid **151** is very flexible to enable changes in the number of fuel cell modules **155** and number of battery sub-packs **154** to be made and therefore adjust the overall battery energy storage capacity. The overall architecture enables multiple independent fuel cell modules **155** to be simultaneously used to provide power to the same DC link voltage (labeled VBATT in FIG. 2) to which the power module battery **154** provides power.

(12) The system utilizes an individual DC/DC converter between each of the fuel cell modules **155**. While this converter boosts the module's output voltage, it also allows compensation for phase and output differences, enabling multiple fuel cell modules **155** to reside in parallel, each providing DC-link power capacity. The result is a shared output that can be regulated around voltage, current, and/or power requirements to meet the needs of the system. For example, it can provide a steady and consistent high voltage output between 400 and 1500 V, such as between 800 and 1500 V or 1200 and 1500 V and a total power output of greater than 1 MW, for example 2, 2.5 or 3 MW. This is the type of voltage and power output that is consistent with a large industrial application like a heavy-duty haul truck. By contrast, a single state of the art fuel cell module typically operates at less than 400 V with an output in the region of 100 KW. This alone would be too low to support a heavy-duty haul truck application. It is only through a configuration like that shown in FIG. 2 and described above that this type of power and voltage output is achievable.

(13) The power module **150** of FIG. 1 replaces the haul truck's conventional diesel generator and

provides the electrical drivetrain with a constant high-voltage DC power supply to facilitate the truck's drive, dump, and auxiliary support functions. The power module **150** resides within the volumetric and configurational constraints of the diesel generator that it replaces within the haul truck system. In addition, the power module **150** structurally interfaces with the truck using the same skid configuration as the diesel generator, allowing the new power module **150** to assemble, integrate, and attach to the truck in a manner similar to the diesel generator.

(14) The integrated power module configuration is one of the preferred embodiments of the present invention because it integrates all of the power generation and power management functions into a single system package, allowing the system to be fully characterized and commissioned without the need of the larger haul truck system. It also minimizes resistive power line losses from high-power harnesses in comparison to other distributed configurations, which would all require longer cable harness lengths resulting in decreased system efficiency. Further, the integrated multi-zone thermal radiator **152** more efficiently manages the varying thermal needs of the power module components by providing a high local and well-connected thermal sink. Further, it utilizes existing primary structural interface and installation aids, requiring no new structural or configurational changes to the haul truck to enable module integration. This is an important aspect so that the truck systems can operate efficiently using the new power source **150** without a substantial reconfiguration of the components on the truck.

(15) FIG. 3 shows in schematic form, the operation of a truck fitted with the power module **150** of the present invention. The numbers shown for the energy balances between the fuel cells **155** and the batteries **154** are for example only and are not intended to be limiting. Each truck begins at position 1 with full tank of hydrogen. It will be appreciated that the tank of hydrogen may comprise several hydrogen storage tanks which, combined, constitute the hydrogen load of the truck. When the truck drives uphill (position 2) it will drain energy from both the fuel cell **155** and the battery **154**. When the truck reaches a flat section (position 3) the fuel cell **155** recharges the battery **154** while also powering the truck. The fuel cell **155** is therefore still using up energy, but some of the energy of the battery **154** used in the uphill section is restored. On a downhill section (position 4), no energy is drawn from the fuel cell **155** and the battery **154** recharges further through the regeneration of energy via the braking system of the truck and the conversion of kinetic energy into stored electrical energy in the battery **154**.

(16) This cycle (positions 2, 3 and 4) can be repeated several times depending on the size and capacity of the hydrogen tank or tanks, and the environment in which the truck is being used and the relative amount of uphill, flat and downhill travel. In the example of possible operation and the cycling of energy from the hydrogen tanks to the battery **154** shown in FIG. 3, the battery **154** is always fully recharged in each cycle by the combination of energy from the hydrogen fuel cell **155** and regenerative energy from downhill sections. The cycle can be repeated 9 times before the hydrogen tank is emptied (position 6) and has to be refueled (position 7). Typically, this refueling may take 15 minutes or less which is significantly quicker than the time it would take to remove and replace empty batteries **154** (which are heavy), or the time it would take to recharge the batteries **154** while still retained on the truck (may often require several hours to fully recharge the batteries **154** once they are drained).

(17) A graph of the possible use and transfer of energy between the battery **154** and fuel cell **155** in a hybrid system is shown in FIG. 4. As discussed above, on an uphill section, the battery is drained of energy, but this is then replaced when the truck is moving in a flat section where the fuel cell **155** is used to recharge the battery **154** and when the truck is moving in the downhill sections where both recharging by the fuel cell **155** and regenerative charging take place.

(18) In mine environments comprising mines and plants, there is an opportunity to significantly reduce the carbon footprint of the operation while at the same time improving operation of the plant. Currently, the operation of the plant will be powered exclusively by electricity from the local grid. Generally, this electricity will have been obtained from fossil fuels and their treatment at a

power plant. Grid systems can be expensive and unreliable as the energy provided can be significantly affected by other users in the grid. Trucks are also powered by fossil fuel derived energy, in most cases by the use of diesel in a diesel generator.

(19) In the present invention, there may be a substantial improvement to the carbon footprint through the use of solar energy to provide power for the plant. This may provide some or all of the power requirements for the plant, and may be topped up as necessary from the grid. This reduces the reliance on the grid and increases security of supply and also decreases the carbon footprint of the plant. Excess solar energy (beyond that used by the plant) is used to produce hydrogen, for example through electrolysis, and this hydrogen is then used in the hybrid system on the trucks as described above. This use of hydrogen in hydrogen fuel cells avoids the need for any diesel on the trucks leading to a significant reduction in greenhouse gas emissions.

Claims

1. A method for powering an ultra heavy-duty truck in a mine environment, in which the ultra heavy-duty truck is retrofitted with an integrated power module, the integrated power module including: a skid coupled to an existing structural interface of the ultra heavy-duty truck, a plurality of hydrogen fuel cell modules coupled to and supported by the skid, a plurality of battery sub packs coupled to and supported by the skid, and integrated power electronics mechanically coupled to and supported by the skid and electrically coupled to the plurality of hydrogen fuel cell modules and the plurality of battery sub packs, the integrated power electronics configured to provide to an electrical drive train of the ultra heavy-duty truck a constant, high voltage DC electrical output, the integrated power module sized and configured to reside within volumetric and configurational constraints of a diesel generator otherwise physically coupled to the structural interface and electrically coupled to the electrical drive train, the method comprising: loading hydrogen on to the ultra heavy-duty truck; using the plurality of hydrogen fuel cell modules to generate electrical power from the hydrogen; supplying at least a portion of the generated electrical power to the electrical drive train of the ultra heavy-duty truck to electrically run the ultra heavy-duty truck; and supplying at least a portion of the generated electrical power to the plurality of battery sub packs to recharge the battery sub packs.

2. The method of claim 1, wherein the plurality of hydrogen fuel cell modules in the ultra heavy-duty truck are arranged in parallel, and wherein the supplying at least the portion of the generated electrical power to the electrical drive train includes simultaneously using the plurality of parallel-arranged hydrogen fuel cell modules to provide electrical power to a single DC link voltage to provide the electrical power to the electrical drive train of the ultra heavy-duty truck.

3. The method of claim 1, further comprising: recharging the plurality of battery sub packs in the ultra heavy-duty truck by taking energy from regenerative braking of the ultra heavy-duty truck.

4. A method for reducing the carbon footprint of a mining environment, in which the mining environment includes a plant and a mine in which material is transported between the mine and the plant by a plurality of ultra heavy-duty trucks, each ultra heavy-duty truck from the plurality of ultra heavy-duty trucks having an integrated power module, the integrated power module including: a skid coupled to an existing structural interface of the ultra heavy-duty truck, a plurality of hydrogen fuel cell modules coupled to and supported by the skid, a plurality of battery sub packs coupled to and supported by the skid, and integrated power electronics mechanically coupled to and supported by the skid and electrically coupled to the plurality of hydrogen fuel cell modules and the plurality of battery sub packs, the integrated power electronics configured to provide to an electrical drive train of the ultra heavy-duty truck a constant, high voltage DC electrical output, the integrated power module sized and configured to reside within volumetric and configurational constraints of a diesel generator otherwise physically coupled to the structural interface and electrically coupled to the electrical drive train, the method comprising: providing locally generated

renewable energy; using the renewable energy to synthesise hydrogen; storing the hydrogen; loading a portion of the stored hydrogen on to each ultra heavy-duty truck from the plurality of ultra heavy-duty trucks; and generating electrical power in the plurality of hydrogen fuel cell modules, the electrical power being used to operate the truck and to recharge the plurality of battery sub packs.

5. The method of claim 4, wherein the locally generated renewable energy is obtained directly from a local electrical grid or is captured locally through one or more of a solar, wind, hydroelectric, geothermal, or nuclear source.

6. A method for providing an integrated power module for an ultra heavy-duty truck, the method comprising: mounting a plurality of hydrogen fuel cell modules to a skid; mounting a plurality of battery sub packs to the skid; mounting integrated power electronics to the skid; electrically connecting the integrated power electronics to each of the hydrogen fuel cell modules and the plurality of the battery sub packs; physically coupling the skid to an existing structural interface of the ultra heavy-duty truck such that the integrated power module resides within volumetric and configurational constraints of a diesel generator otherwise physically coupled to the structural interface and electrically coupled to an electrical drive train of the ultra heavy-duty truck; and electrically coupling the integrated power electronics to the electrical drive train of the ultra heavy-duty truck.

7. The method of claim 6, wherein the integrated power electronics are configured to provide the electrical drive train a constant, high voltage DC electrical output with a power of at least 0.5 MW and a voltage between 400 V and 1500 V.

8. The method of claim 7, wherein the integrated power electronics include an individual DC/DC converter coupled to an output of each hydrogen fuel cell module and configured to boost the output voltage thereof, the DC/DC converters collectively operable to compensate for phases and output differences between the plurality of hydrogen fuel cell modules and to enable a shared output to be regulated to allow the integrated power electronics to provide the constant, high voltage DC electrical output.

9. The method of claim 7, wherein the electrical output has a power of at least 1 MW.

10. The method of claim 7, wherein the energy storage capacity of the power module is in excess of 1 MWh.

11. The method of claim 6, further comprising: mounting a cooling system to the skid, the cooling system configured to cool the plurality of hydrogen fuel cell modules and the plurality of battery sub packs.

12. The method of claim 11, wherein the mounting the cooling system to the skid includes mounting a multi-zone radiator with integrated fans and coolant fluid loops to the skid.

13. The method of claim 6, further comprising: electrically connecting the plurality of hydrogen fuel cell modules in parallel; and electrically connecting the integrated power electronics to a single DC link voltage.

14. The method of claim 6, wherein the integrated power module is retrofitted into the ultra heavy-duty truck, the ultra heavy duty truck having the diesel generator physically coupled to the structural interface and electrically coupled to the electrical drive train, the method further comprising: electrically disconnecting the diesel generator from the electrical drive train; and physically removing the diesel generator from the structural interface after electrically disconnecting the diesel generator and prior to physically coupling the skid to the structural interface.
